

Jennifer Chinyere

AI/ML Engineer | Backend Engineer | Data Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer and Full-Stack Software Engineer with experience designing and deploying intelligent platforms powered by LLMs, computer vision, OCR, and speech recognition. Built production-style AI systems using FastAPI, React, PostgreSQL, AWS, and modern cloud infrastructure, including an AI-powered clinical referral platform that reduced referral processing time from 27 minutes to 2 minutes and a computer vision system that improved radiologist throughput by 2.1x. Passionate about building reliable AI products that combine scalable backend engineering, machine learning, and intuitive user experiences to drive measurable impact.

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, HTML, CSS, Bash, SQL, C++
- Frameworks: FastAPI, Django, React, Next.js, Node.js, SQLAlchemy, Pydantic, Tailwind CSS, TanStack Query
- AI/ML: OpenAI API, Groq Llama, Whisper, Tesseract OCR, Scikit-learn, spaCy, MLflow, Ragas, ChromaDB, LiteLLM, YOLOv8, PyTorch, OpenCV, Grad-CAM, NumPy, Pandas, NLTK, Jupyter Notebook
- Databases & Cloud: PostgreSQL, MongoDB, Redis, AWS (S3, EC2, Lambda), Render, Vercel, Railway, Supabase
- AI Developer Tools: Cursor, GitHub Copilot, Claude Code, ChatGPT, Gemini, Windsurf
- DevOps & Tools: Docker, Git, GitHub Actions, CI/CD, REST APIs, Linux, PowerShell, Postman, Jira, Slack, Zapier
- Testing & Security: Pytest, HTTPX, Jest, Cypress, React Testing Library, JWT, OAuth2, RBAC

TECHNICAL PROJECTS

MediFlow - AI Healthcare Referral Management Platform

FastAPI, PostgreSQL, React, TypeScript, Docker, AWS, OpenAI API, Groq Llama 3.1, Whisper, Tesseract OCR, Scikit-learn(TF-IDF, Cosine Similarity), WebSockets

- Architected a multi-tenant healthcare referral platform with 50+ RESTful APIs using FastAPI, PostgreSQL, React, TypeScript, AWS, and Docker, streamlining referral intake and clinical workflows across multiple facilities.
- Built an AI-powered document intelligence pipeline using Tesseract OCR, Whisper, and Groq Llama 3.1 to automate document extraction, speech transcription, clinical summarization, and AI-assisted referral reason classification, reducing processing time from 27 minutes to 2 minutes with 92% OCR accuracy.
- Implemented duplicate patient detection using Scikit-learn and fuzzy matching techniques, alongside JWT authentication, RBAC, WebSockets, and DICOM-ready storage, contributing to a projected 45% reduction in referral handling time and approximately 360 clinical hours saved monthly per clinic.

OsteoDetect - AI-Powered Bone Fracture Detection System

Python, FastAPI, PyTorch, YOLOv8, OpenCV, Grad-CAM, HTML5, CSS3, Docker, Vercel, Railway

- Engineered an AI-assisted fracture detection platform using YOLOv8, FastAPI, and OpenCV, enabling automated fracture detection from X-ray images with inference times of approximately 3 seconds per radiograph.
- Integrated Grad-CAM explainability visualizations and confidence scoring to improve clinical interpretability, contributing to an estimated 85% clinician confidence boost during evaluation.
- Built RESTful inference APIs and an end-to-end imaging workflow enabling clinicians to upload radiographs, receive AI-assisted predictions, and visualize fracture localization through an intuitive web interface.
- Optimized diagnostic workflows through AI-assisted triage, reducing manual review bottlenecks, saving an estimated 40–50 minutes per radiologist per shift, and 82% user satisfaction during testing.

EDUCATION

Diploma in Computer Science & Engineering
Akirachix

2024
CGPA: 9.06/10